3.3. Weighted permutation entropy

By plotting the signals of the glucose concentration for healty and diabetic patients, we observed that the signal amplitude is bigger for the diabetic group. We then tried to compute the WHXC feature plan which stand for Weighted HxC feature plan, using the weighted permutation entropy as defined in **"Weighted-permutation entropy: A complexity measure for time series incorporating amplitude information"** by B. Fadlallah, B. Chen, A. Keil and J. Príncipe.

As we used a new definition of the permutation entropy, we had to find new (m,τ) parameters that maximize the distance between the two classes:

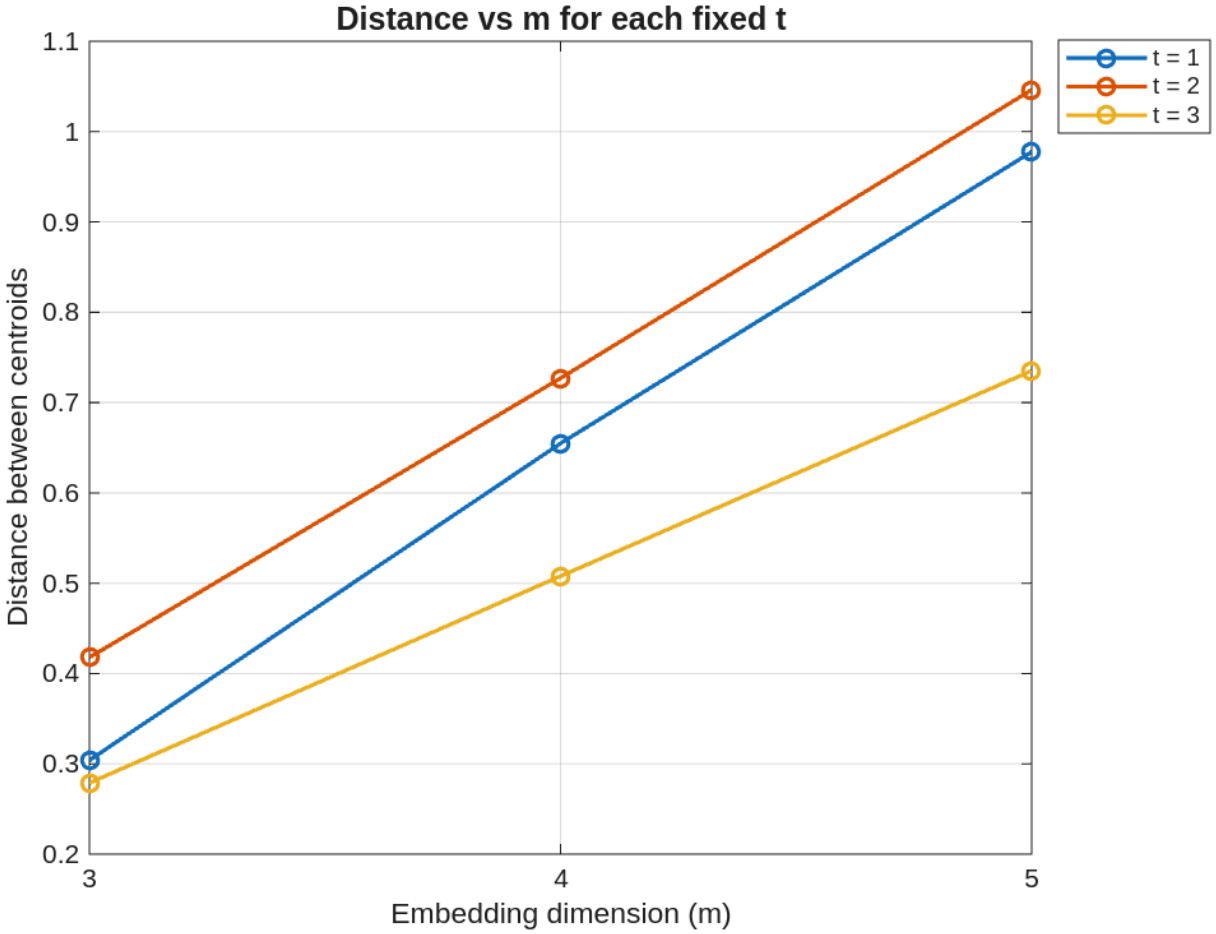


Figure 9: Weighted HxC feature plan of the centroids of the diabetic dataset

As shown, the two best sets of parameters seem to be (m = 5, τ = 2) and (m = 4, τ = 2):

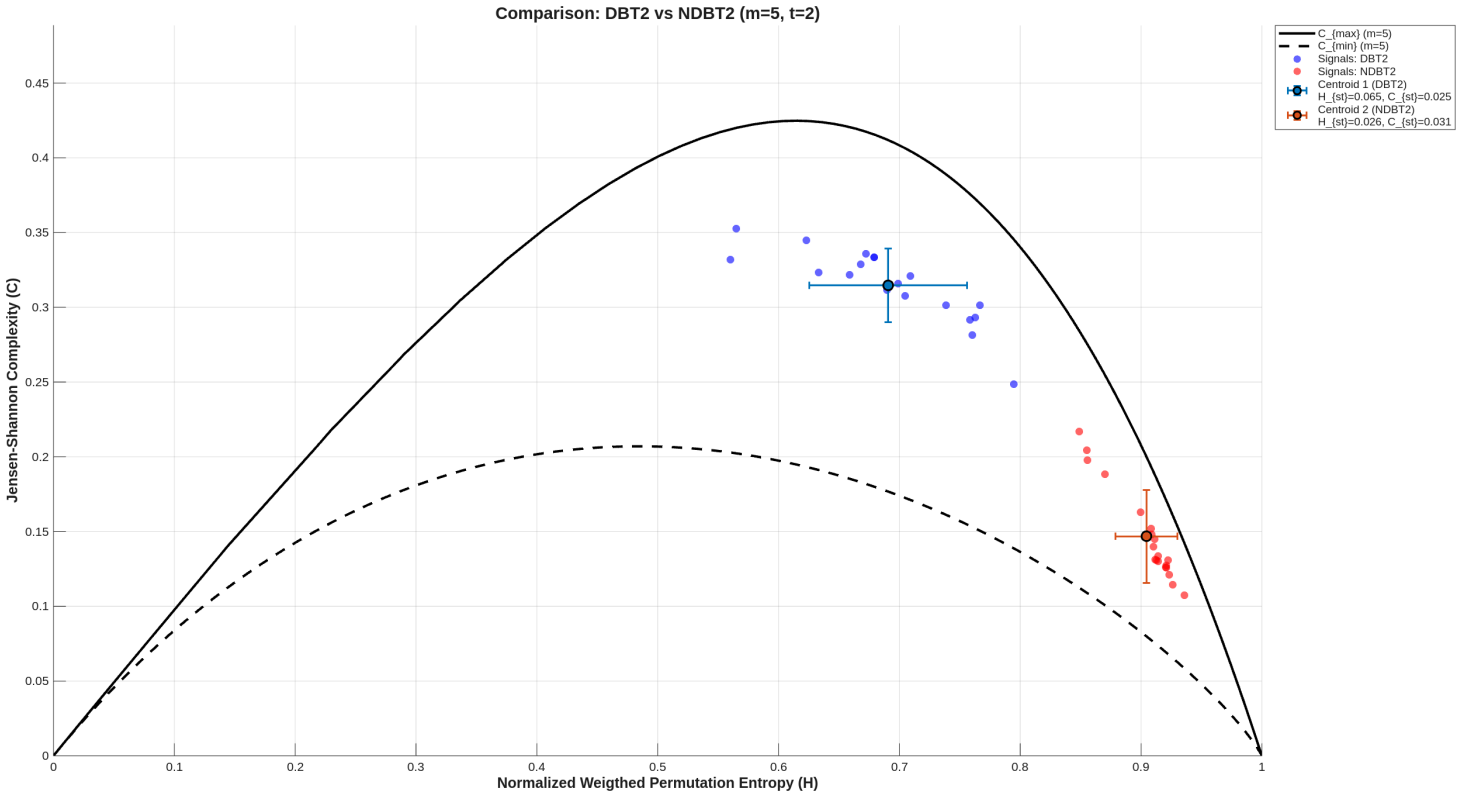


Figure 10: Comparison of the weighted permutation entropy for the m = 5 and $\tau = 2$

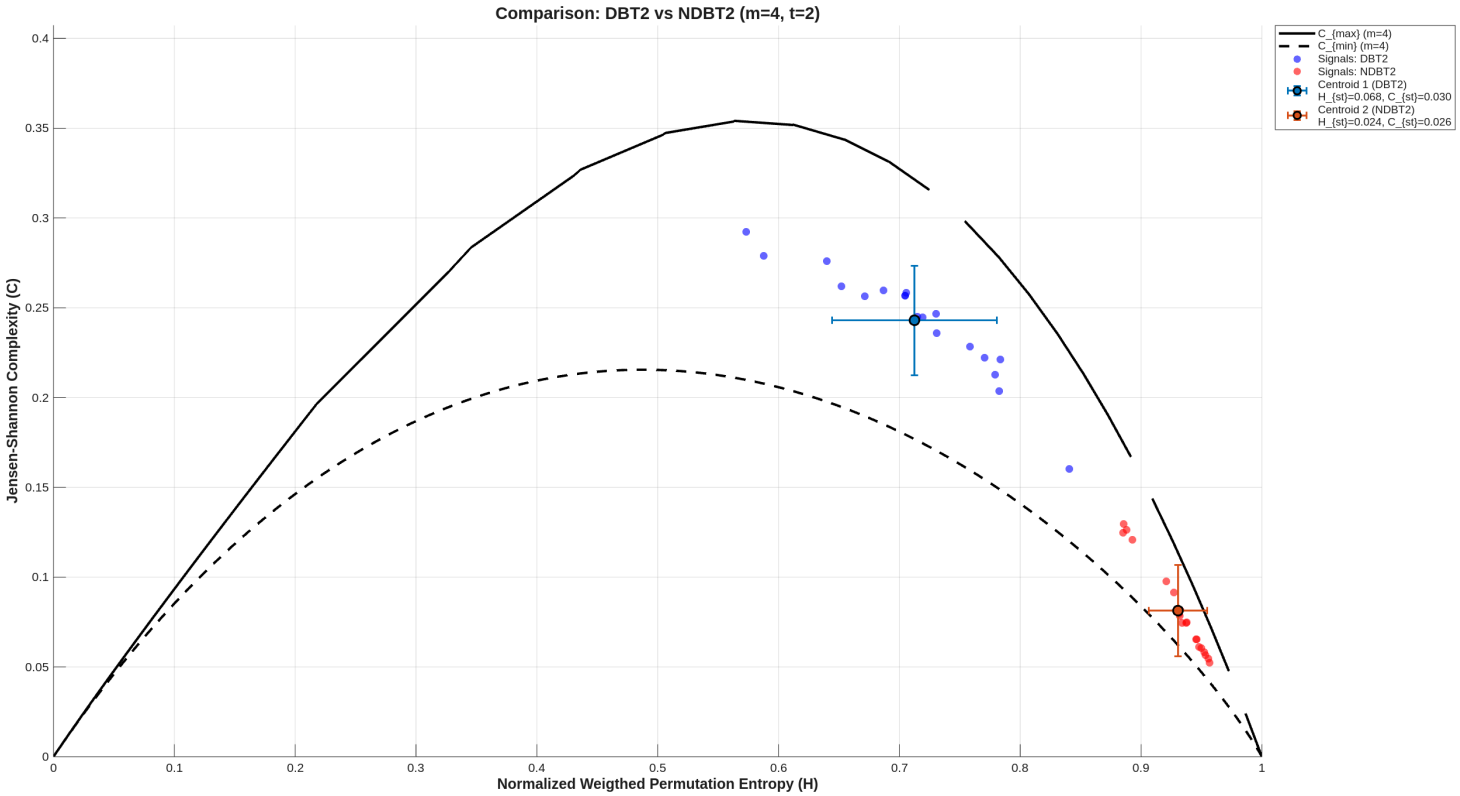


Figure 11: Comparison of the weighted permutation entropy for the m = 4 and $\tau = 2$

Nb : discontinuity in the theoritical max complexity are due to MATLAB overlapping plots.

Here we observe an interesting behavior: (5,2) configuration gives the maximum distance between the two classes of patients but the (4,2) configuration shows a more satisfying space distribution. By more satisfying, we mean that the two clouds of points respectively lie in areas consistent with their signal type (a changing mix between random and chaotic).

Result : For the rest of the study, we will use (m = 4, τ = 2) set of parameters to compute the weighted permutation entropy and its associated complexity, as it better reflects the nature of the signals.

NB : This criteria of computing the euclidean distance between the centroids to find the best set of parameters seems a bit too naive if used alone. We might include the standard deviation but maybe also some thresholds to ensure each centroid's group stays in an area consistent with their physical behavior.

What to do next? Try to compute the mutual information and transfer entropy using this type WPE?

3.4. Lyapunov exponent

In order to find others metrics to feed the feature space of the classifier, we also computed the average lyapnuov exponent of each class, using the glucose concentration:

Group of patient	Mean Lyapunov exponent	Standard deviation
Diabetic	0.000278	0.000162
Non-diabetic	0.000203	0.000123

Table 3: Mean Lyapunov exponent for each group of patient with the samping frequency $f_s = 1/900$ Hz

Result : At first sight, it seems that the lyapunov exponent is in this case not a relevant metric to distinguish the two classes. However, the standard deviation is too high for now to really conclude anything.

NB : How to better the standard deviation? Change the Lyapunov matlab function?

3.5. Higuchi's fractal dimension

We also computed the fractal dimension of the glucose concentration using Higuchi's method. The question then is which depth k_{max} to use. Here for $k_{max} = 2$, corresponding to a evolution of 30min:

Group of patient	Mean Fractal Dimension	Standard deviation
Diabetic	1.1540	0.0453
Non-diabetic	1.2737	0.0124

Table 4: Mean Fractal Dimension for each group of patient with the max depth $k_{max} = 2$

The time of observation seemed a bit too short to distinguish different behaviors. We tried bigger time windows:

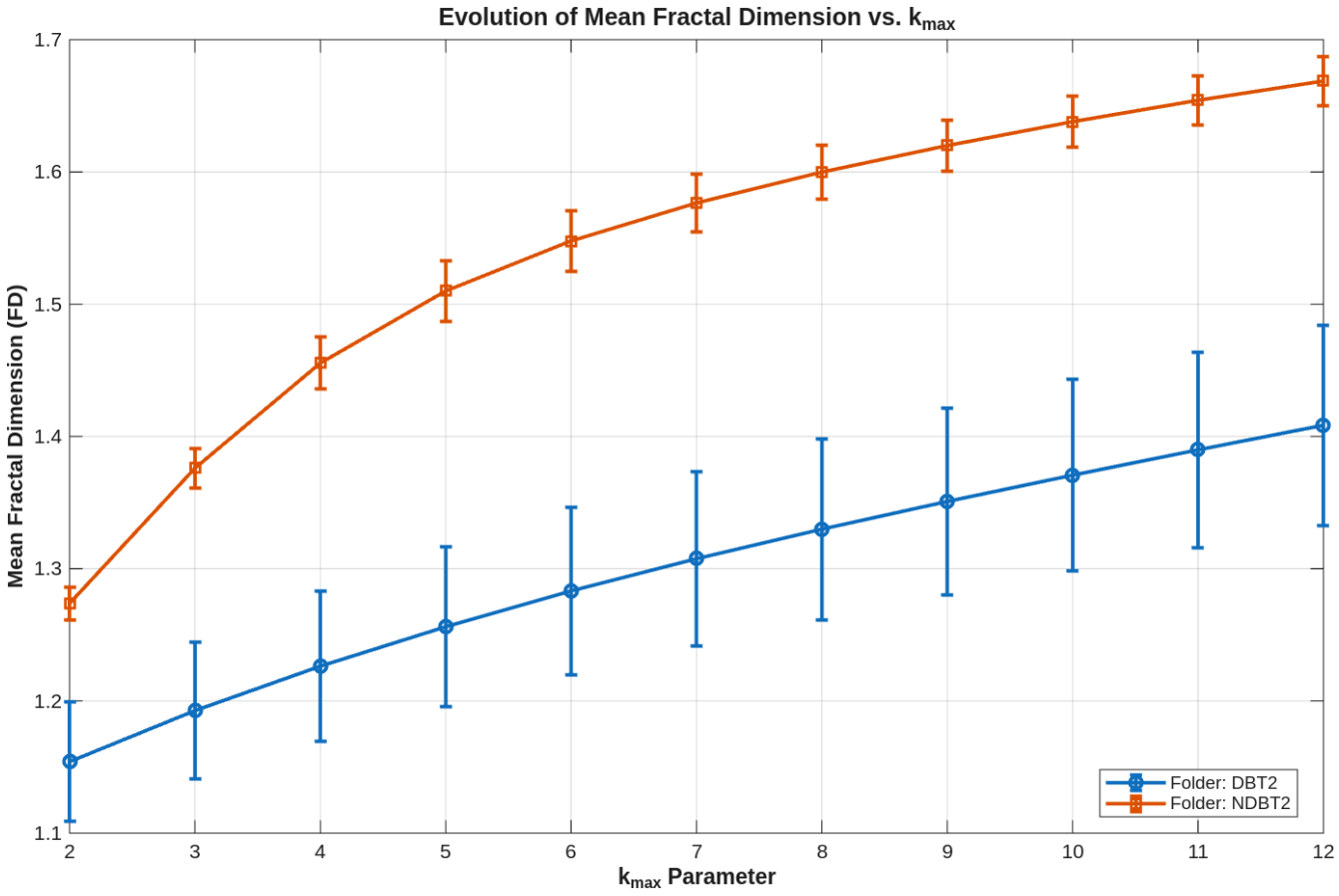


Figure 12: Mean Fractal Dimension for each group of patient with different max depth k_{max}

Result : After $k_{max} = 6$, which corresponds to a time window of 1h30, we seems to have reached a stationnary difference between diabetic and non-diabetic patients of around 0.2603.

NB : The standard deviations are not overlapping but is this difference enough to be used in the feature set?

4. To try next

4.1. Multi-dimensional

- 2-D Permutation Entropy (see EntropyHUB)

4.2. Mono-dimensional

On the glucose concentration ?

- Fourier Spectrum
- AUC (power density)